Toxicity Reduction: All methods substantially reduce toxicity from baseline (0.23). Classifier-based DPO achieves 0.11, while probe-based methods reach 0.14-0.15. Probe-based DPO produces 27-36% more toxic outputs than the classifier baseline, but still delivers 35-40% improvement over baseline.
Probe Detectability: Training against probes preserves detectability better than training against classifiers—a surprising reversal. Classifier-based DPO degrades probe accuracy substantially: held-out probes drop to 0.866 AUC, retrained probes to 0.770. In contrast, single-probe DPO maintains 0.938 (held-out) and 0.926 (retrained), while 10-probe DPO achieves 0.994 and 0.992, respectively. Critically, training probes themselves retain 0.977-0.987 AUC, indicating the model doesn't evade its training signal through Goodhart-style obfuscation. We hypothesize probe-based DPO may preserve detectability because the preference learning objective requires discriminating between chosen and rejected responses based on probe scores. This creates incentives to maintain—rather than destroy—the representational features probes detect. Classifier-based DPO lacks this incentive since the model never sees the representations underlying classifier judgments.

Using 10 probes versus 1 improves held-out probe AUC from 0.938 to 0.994 (90% error reduction). Whether this justifies a 10× computational cost depends on application requirements.

General capabilities remain intact. WikiText-2 perplexity hovers around 37 for all methods (Table 1). Training distribution perplexity increases from 39 to 41 across all methods—an expected DPO side-effect where the KL penalty pushes the policy away from the reference distribution.

5.2 SFT

We present results from three SFT conditions: baseline (standard SFT without probe regularization), single probe-based regularization, and 10 probe-based regularization. Table 2 shows final metrics, while Figure 3 displays training dynamics.

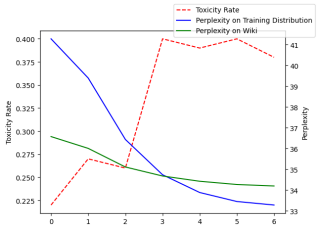
Standard SFT on toxic prompts increases toxicity from 0.22 (base model) to 0.38, as the model overfits to the toxic training distribution. Probe-based regularization partially mitigates this increase, with both single-probe and 10-probe training reaching 0.35—still 59% more toxic than the base model, but representing an 8% relative reduction compared to standard SFT. These modest improvements contrast sharply with DPO results (0.11-0.15 toxicity). This aligns with SFT's intended use case from Section 3.3: preventing increases in toxicity when training on unfiltered data, rather than actively detoxifying the model.

Unlike DPO, SFT does not show a clear advantage for probe-based methods in preserving detectability. All methods maintain high probe AUC across evaluation types. Baseline SFT achieves 0.974 (training probes), 0.957 (held-out), and 0.958 (retrained). Probe-based methods show modest degradation in training probe AUC (0.942-0.958), suggesting the model learns to partially evade the specific probes used during training. However, held-out probes (0.943-0.947) and especially retrained probes (0.957-0.973) maintain relatively high accuracy. This contrasts with DPO, where probe-based methods dramatically outperformed the baseline in preserving detectability. We speculate this difference arises because SFT's gradient-based regularization allows direct evasion of probe features, while DPO's preference learning objective incentivizes maintaining those features for discrimination. Importantly, retraining probes after optimization recovers detection accuracy to near-baseline levels (0.957-0.973). This indicates toxicity remains linearly separable in representation space after training, just along slightly different directions. This result, consistent with DPO findings, suggests probe ensembles are unnecessary when retraining is feasible. Training against probes reduces their AUC, but not by much. We believe this is because fine-tuning pushes the probe scores to become smaller, but the relative ordering of toxic/non-toxic examples stays the same. Thus, refitting the threshold to a lower value restores much of the accuracy.

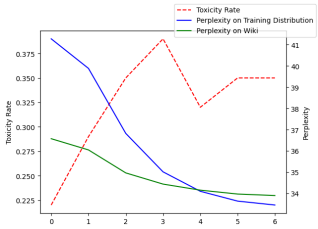
Probe regularization doesn't impede language modeling. All methods achieve similar reductions in training perplexity (top panels of Figure 3), indicating the model successfully learns from the training data despite the probe penalty. WikiText-2 perplexity remains around 34 across all conditions (Table 2), confirming general capabilities are unaffected. Multi-probe ensembles provide no advantage, with both single and 10-probe methods achieving identical toxicity rates (0.35) and comparable probe AUCs.

	Toxicity Rate	AUC train probes	AUC held-out probes	AUC retrained probes	Perplexity Training Distribution	Perplexity Wiki
Classifier	0.38	0.974	0.957	0.958	33.279	34.195
1 probe	0.35	0.942	0.947	0.973	33.448	33.895
10 probes	0.35	0.958	0.943	0.957	33.566	33.997

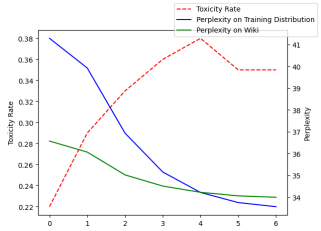
Table 2: Final results of SFT training.



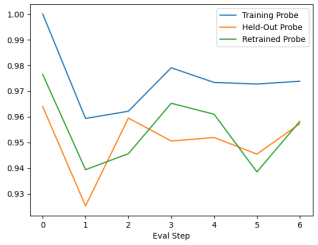
(a) Toxicity rate, perplexity on training distribution, and perplexity on the Wiki-dataset during SFT training using a **text-based toxicity classifier**.



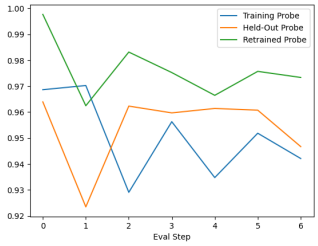
(b) Toxicity rate, perplexity on training distribution, and perplexity on the Wiki-dataset during SFT training using **one probe-based classifier.**



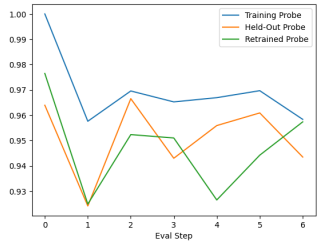
(c) Toxicity rate, perplexity on training distribution, and perplexity on the Wiki-dataset during SFT training using **10 probe-based classifiers**.



(d) Area Under Curve (AUC) of held-out and newly trained probes during SFT training using a **text-based toxicity classifier**.



(e) Area Under Curve (AUC) of the probe used for training, held-out and newly trained probes during SFT training using **one probe-based classifier**.



(f) Area Under Curve (AUC) of probes used for training, held-out, and newly trained probes during SFT training using **10 probe-based classifiers**.

Figure 3: Changes in probe accuracy, toxicity rat,e and model capability when training with SFT using a text-based classifier, one probe-based classifier, or 10 probe-based classifiers.

6 Conclusion

We investigated whether training language models against internal representations inevitably leads to probe evasion—a central concern for interpretability-based alignment methods. Using toxicity reduction as a testbed, we compared probe-based and classifier-based training across supervised fine-tuning and direct preference optimization.

Our main finding is that the training method determines probe viability. Probe-based DPO preserves detectability substantially better than classifier-based DPO, maintaining high held-out probe accuracy while the baseline shows severe degradation. We hypothesize this stems from DPO's preference learning objective, which incentivizes maintaining—rather than obfuscating—the representational features that distinguish chosen from rejected responses. This suggests that optimizing against internal representations can strengthen rather than destroy the monitoring signal when the training objective aligns with probe detection.

In contrast, probe-based SFT shows no such advantage. While probe regularization modestly reduces toxicity increase during training on toxic data, it neither outperforms baseline methods nor prevents partial probe evasion. This method-specific difference suggests probe-based training is most viable for preference-based objectives rather than gradient-based regularization.

Our second finding limits the practical need for probe diversity. Multi-probe ensembles offer marginal improvements over single probes in both training methods. More importantly, probes retrained after optimization consistently recover high detection accuracy, indicating the targeted behavior remains linearly separable in representation space despite training pressure. When probe retraining is feasible, held-out ensembles provide little additional value.

These results should be interpreted alongside concurrent work showing that architectural robustness (attention-based and max-pooling probes) offers a more scalable solution to probe evasion Gupta and Jenner [2025], and that probe-based preference learning outcomes are highly sensitive to implementation details Cundy and Gleave [2025]. Our contribution delineates when standard linear probes remain viable monitoring tools: specifically, when using preference-based objectives or when probe retraining is available.

Limitations. Our findings are limited to toxicity detection at 1B model scale. Toxicity is easily detectable from model outputs, making it a conservative testbed—probe-based training should show larger advantages for behaviors like deception or reward hacking where output-based detection fails. The substantial metric noise during training raises questions about result stability and suggests the need for multiple training runs with larger evaluation sets. Most critically, we cannot determine whether probe-based DPO genuinely eliminates toxic reasoning or merely maintains detectable toxic representations while producing non-toxic outputs.

Future directions. The key question is whether probe-based DPO's detectability advantage extends to behaviors that are difficult to detect from outputs alone—precisely the scenarios where probe-based training should prove most valuable. Testing on deception, sycophancy, and internal reward hacking would clarify whether this approach offers practical benefits beyond output-based training. Additionally, investigating why DPO preserves detectability while SFT does not could inform the design of more robust interpretability-based training methods.

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